Open-World Explained-Rarity Filtering for Health-Insurance Claim Fraud: Problem Formulation, Benchmark Design and Preliminary Synthetic Evidence

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***Abstract*—Health-insurance fraud detectors trained on known fraud categories can force novel schemes into familiar classes and confuse clinically rare but legitimate claims with fraud. We formulate *explained-rarity filtering*: a claim is escalated only when it is unfamiliar and unexplained by clinical and provider evidence. We report a tabular prototype, independent and controlled deep-model comparisons, a temporal relation-summary pilot, and a separate custom typed-node temporal graph baseline evaluated over 12 synthetic folds (four held-out mechanisms and three generator seeds). The graph baseline achieved mean known-class accuracy 0.954 ± 0.012 and macro-F1 0.645 ± 0.107; novel-mechanism AUROC was 0.732 ± 0.170, while recall at the nominal $\alpha = 0.05$ rejection threshold was only 0.180 ± 0.200. Performance varied sharply by mechanism: collusion AUROC was 0.494 ± 0.049, close to chance. The graph encoder is custom PyTorch R-GCN/HGT-style message passing, not canonical PyG HGTConv/RGCNConv, and uses synthetic Admission/Diagnosis proxies. No experiment uses real claim-level fraud labels. We report the measured outcomes and remaining limitations without claiming real-world performance.**

***Index Terms*—health-insurance fraud, open-world recognition, conformal prediction, anomaly detection, graph neural networks**

I. Introduction

Large claim volumes in health-insurance systems, including IRDAI-regulated insurance and the AB-PMJAY scheme in India, make exhaustive manual review impractical. Fraud mechanisms such as upcoding, unbundling, phantom billing and collusive referral rings also change over time. Three problems follow.

1) **Closed-world classification.** A softmax classifier assigns a new scheme to the nearest known class, often with high confidence.
2) **Rarity is not fraud.** Pure anomaly scores flag legitimate severe cases because they are rare, complex and expensive.
3) **Alert fatigue.** Raw scores carry no statistical meaning, so a fixed threshold drifts and review budgets are exceeded.

Contributions. (1) A problem formulation, *explained-rarity filtering*, that separates unfamiliarity from unexplainedness. (2) A pipeline (OW-ERF) combining post-hoc novelty fusion, split-conformal rejection and a heteroscedastic clinical expectation filter. (3) A benchmark protocol with a synthetic claim generator, leave-one-mechanism-out evaluation and leakage control. (4) Preliminary results from tabular, deep-model, temporal relation-summary and typed-node temporal graph experiments, including mechanism-specific failure cases.

II. Related Work

Open-set and open-world recognition. Open-set recognition [1], [2] and out-of-distribution (OOD) baselines [3]–[5] score unfamiliarity from logits or embeddings. On graphs, OpenWGL [6] rejects unseen-class nodes with a variational graph autoencoder, and GNNSafe [7] applies energy-based detection to graph neural networks.

Graph fraud and anomaly detection. CARE-GNN [8] counters camouflage with reinforcement-learned neighbor selection, and BWGNN [9] addresses heterophily. DOMINANT [10] and the PyGOD library [11] cover reconstruction-based detection, while BOND [12] and GADBench [13] provide benchmarks. These methods output novelty or anomaly scores and do not separate rare-but-clinically-explained claims from unexplained ones.

Conformal prediction. Split-conformal methods [14]–[16] give finite-sample marginal guarantees under exchangeability. Barber et al. [17] study relaxations, and Huang et al. [18] apply conformal prediction to graph neural networks. Temporal claim streams violate exchangeability, so our guarantee is approximate and is tested empirically.

Heteroscedastic regression. Predicting a mean and a variance under a Gaussian likelihood [19], [20] lets the expected spread grow with severity.

Healthcare fraud. Prior work applies supervised and unsupervised learning to Medicare-style data [21].

III. Problem Formulation

Let a claim v have features x_v, a graph neighborhood, and a label y_v from benign, known mechanisms, or novel mechanisms. Training sees only benign and known mechanisms. At test time a claim is *unfamiliar* if its conformal p-value satisfies $p_v \le \alpha$, and *unexplained* if its clinical inconsistency ratio a_v exceeds a calibrated threshold τ. A claim is escalated if and only if it is both.

To prevent temporal leakage, the embedding of claim v at adjudication time t uses only the causal neighborhood

$$\mathcal{N}_r(v;t) = \{u : (u,v) \in E_r,\ t(u,v) \le t\}. \tag{1}$$

IV. Method

Fig. 1 shows the pipeline.

A. Encoder

The target architecture specifies a timestamped heterogeneous graph of claims, beneficiaries, hospitals, doctors, admissions, packages and diagnoses, with time-aware edges and an R-GCN or HGT encoder using PyTorch Geometric [22]–[24]; the earlier relation-summary pilot did not implement this arbitrary-node PyG encoder. A separate dependency-light typed-node temporal message-passing encoder has since been evaluated over 12 synthetic folds; those measured results are reported in the typed-node temporal graph benchmark subsection below. It is a custom PyTorch R-GCN/HGT-style implementation rather than the canonical PyG HGTConv/RGCNConv. Three complementary branches are separately implemented and trained in the released synthetic benchmark: an LSTM over recent claim-history sequences, a 1D-CNN over ordered procedure-code tokens, and a Transformer over relational-context tokens. The standalone experiment compares each architecture independently rather than claiming that it is already fused with a full HGT/R-GCN. The three models use different input modalities by design, so the comparison is a comparison of architecture-plus-modality pipelines. A separate relation-summary pilot implements causal relation-specific attention and additional OW-ERF stages, but it is not a full arbitrary-node heterogeneous graph encoder.

B. Novelty score

From classifier logits z, the energy is

$$E(x) = -T \log \sum_k \exp(z_k/T). \tag{2}$$

The Mahalanobis score $M(h)$ is the minimum over known classes of the squared Mahalanobis distance of embedding h to the class mean under the class covariance. With F_E and F_M the empirical CDFs of E and M on training data, the fused score is

$$u = \tfrac{1}{2}F_E(E) + \tfrac{1}{2}F_M(M). \tag{3}$$

C. Conformal rejection

On a calibration set disjoint from training and containing only known-class claims, with n calibration scores u_i, the p-value is

$$p_v = \frac{1 + \#\{i : u_i \ge u_v\}}{n+1}, \tag{4}$$

and a claim is rejected as unfamiliar when $p_v \le \alpha$ with $\alpha = 0.05$.

D. Explained-Rarity Filter

A heteroscedastic model trained on benign claims only predicts a mean μ_j and standard deviation σ_j for $j \in \{\text{log-amount}, \text{length of stay}\}$ from *exogenous* inputs (a severity proxy from prior history, comorbidity count, age) by minimizing

$$\mathcal{L}_g = \frac{1}{N}\sum_i \sum_j \Big[\ln \sigma_{ij} + \frac{(o_{ij}-\mu_{ij})^2}{2\sigma_{ij}^2}\Big]. \tag{5}$$

The inconsistency ratio is

$$a_v = \max_j \max\Big\{0, \frac{o_j - \mu_j}{\sigma_j}\Big\}. \tag{6}$$

The threshold τ is set as a quantile of a_v on benign calibration claims rather than by hand. The expectation model must not condition on fields a fraudster controls, such as the claimed diagnosis code; otherwise upcoding would make fraud look explained.

E. Additional OW-ERF pilot components

The separate relation-summary pilot implements episodic pseudo-novel energy-margin training, energy/Mahalanobis fusion, split-conformal rejection, legitimate-only clinical expectation and structural-reference thresholds, explained-rarity filtering, HDBSCAN grouping (with DBSCAN fallback when HDBSCAN is unavailable), cluster cards and audit-oriented metrics. It uses past-only entity summaries over six relation types, not a PyG HGT/R-GCN over arbitrary typed nodes. A distinct custom typed-node temporal encoder is now available in `owerf_temporal_heterograph.py`; its metrics are intentionally not folded into this pilot's results. The standalone LSTM/CNN/Transformer experiment is a separate benchmark and must not be reported as the full graph model.

V. Experimental Protocol

Data. A synthetic generator produces benign claims, 5% rare-severe legitimate claims, and four mechanisms: upcoding, unbundling, phantom billing, and collusion (represented in the prototype by two proxy features only). The fraud rate is 10%.

Splits and metrics. Three experiments are reported separately. The earlier tabular prototype uses five random seeds and its own i.i.d. split. The standalone deep comparison uses

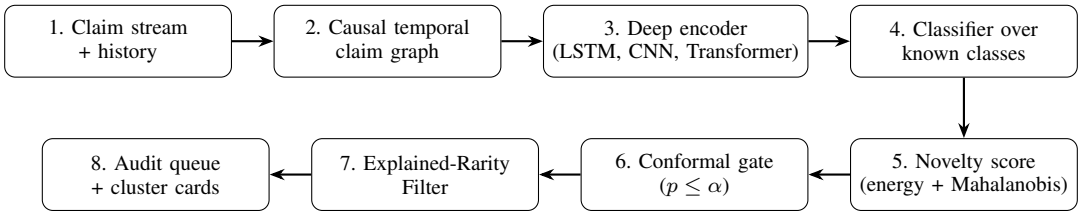


Fig. 1. Pipeline of OW-ERF. Stages 1–4 learn representations; stages 5–8 make the decision.

TABLE I
LEAVE-ONE-MECHANISM-OUT RESULTS (SYNTHETIC DATA, TABULAR PROTOTYPE, MEAN OF 5 SEEDS). RECALL IS ON THE HELD-OUT MECHANISM; "RARE FLAG" IS THE FRACTION OF RARE-SEVERE LEGITIMATE CLAIMS FLAGGED.

Held-out mechanism	Recall conf.	Recall +ERF	Rare flag conf.	Rare flag +ERF
Upcoding	0.574	0.560	0.026	0.006
Unbundling	0.493	0.372	0.020	0.003
Phantom billing	0.496	0.411	0.133	0.026
Collusion (proxy)	0.586	0.211	0.016	0.002

three generator seeds (42, 43, 44), four leave-one-mechanism-out folds, an internal known-only validation subset, a separate known-only calibration split, and an untouched test split, yielding 36 independent model fits. Known-class accuracy is calculated only on test claims from classes observed during training; novel-mechanism AUROC/AP treat the held-out mechanism as positive for novelty scoring. The v0.2 relation-summary pilot uses chronological train, selection, calibration and deployment windows. These results are synthetic, and the standalone benchmark does not establish real-world performance.

Baseline. Isolation Forest [25], with its alert count matched to the number of OW-ERF escalations.

VI. PRELIMINARY RESULTS

Table I retains the earlier tabular-only reference results (means over five seeds). The conformal stage rejects between 4.3% and 4.6% of benign claims, close to the nominal 5% on this i.i.d. synthetic split; this does not demonstrate validity under temporal drift. After the filter, benign escalation falls to about 0.5%. The filter cuts flags on rare-severe legitimate claims most clearly for phantom billing ($13.3\% \rightarrow 2.6\%$). It also lowers recall: for collusion the drop is large ($0.586 \rightarrow 0.211$), because those claims look clinically plausible and would require graph structure that this prototype lacks. At a matched alert count, Isolation Forest recalls only 0.02–0.04 of the three attribute-level mechanisms (0.20 for collusion) while flagging 22%–28% of rare-severe legitimate claims. All of these results are conditional on the generator.

A. Standalone deep-model comparison

Table II reports the mean and standard deviation across the 12 model/fold/seed runs per architecture (four held-out mechanisms by three generator seeds). Known-class accuracy excludes the held-out mechanism; novelty metrics treat the held-out mechanism as novel. This distinction is important because the model may classify known classes reasonably while failing to reject unfamiliar fraud.

The Transformer has the strongest mean known-class accuracy and novel AUROC in this 5-epoch pilot, but the high dispersion and low mean novel recall show that performance is not yet robust. The average known false-rejection rates at $\alpha = 0.05$ were 0.050 (LSTM), 0.047 (1D-CNN), and 0.045 (Transformer); this calibration target applies to known classes, not unseen fraud. The full JSON/CSV and selected checkpoints are generated by the dedicated GitHub Actions workflow. These metrics are synthetic generator results, not real insurance claim results.

B. Controlled architecture comparison with shared inputs

The native standalone comparison above changes both architecture and input modality. To isolate the architecture effect more fairly, we ran a paired controlled comparison: the LSTM, a continuous-input 1D-CNN, and Transformer received the same standardized six-token, 16-channel relational-context tensor and the same padding mask. Scaling was fitted only on valid training tokens; padded positions were zeroed, the LSTM packed valid sequence lengths, the CNN used mask-aware pooling, and the Transformer used its key-padding mask. The experiment used the same four held-out mechanisms and generator seeds 42, 43, and 44, for 36 model fits per mode (72 total). Checkpoints were selected using validation cross-entropy; the test split was not used for model selection. The 25-epoch pilot is preserved in a separate report; Table III presents the completed 60-epoch follow-up.

The large LSTM and CNN known-class-accuracy changes under shared inputs show that the original score gap was substantially confounded by modality: collusion information is encoded in history, unbundling in code tokens, and upcoding/phantom cues in the claim-context view. The Transformer is numerically unchanged because its native input was already that context view. Under shared context, novelty AUROC improves for LSTM and CNN, but dispersion remains high and novel recall at the conformal threshold is still low/variable. The shared-context known false-rejection rates were 0.048 ± 0.019 (LSTM), 0.054 ± 0.015 (1D-CNN), and 0.061 ± 0.021 (Transformer); the Transformer remains slightly above the nominal 5% rate. Validation-selected best epochs reached the 60-epoch cap in 3/12 shared-context LSTM runs and 1/12 shared-context CNN runs (none of the Transformer runs),

TABLE II
INDEPENDENT DEEP-MODEL RESULTS ON THE SYNTHETIC HELD-OUT-MECHANISM BENCHMARK (MEAN ± STANDARD DEVIATION ACROSS 12 FOLDS/SEEDS PER MODEL).

Model	Known accuracy	Known macro-F1	Novel AUROC	Novel AP	Novel recall ($\alpha = 0.05$)
LSTM	0.630 ± 0.183	0.380 ± 0.105	0.576 ± 0.189	0.139 ± 0.277	0.156 ± 0.288
1D-CNN	0.565 ± 0.114	0.279 ± 0.048	0.561 ± 0.126	0.092 ± 0.093	0.120 ± 0.159
Transformer	0.809 ± 0.115	0.646 ± 0.137	0.693 ± 0.142	0.142 ± 0.215	0.156 ± 0.251

TABLE III
CONTROLLED DEEP-MODEL COMPARISON ON SYNTHETIC DATA (60-EPOCH MAXIMUM; MEAN ± STANDARD DEVIATION ACROSS 12 FOLD/SEED RUNS PER MODEL). NATIVE MODE USES EACH MODEL'S INTENDED INPUT; SHARED-CONTEXT MODE GIVES ALL MODELS THE SAME STANDARDIZED TOKENS.

Model	Native acc.	Shared acc.	Native AUROC	Shared AUROC	Shared AP	Shared novel recall ($\alpha = 0.05$)
LSTM	0.519 ± 0.098	0.979 ± 0.010	0.539 ± 0.130	0.721 ± 0.210	0.169 ± 0.105	0.250 ± 0.166
1D-CNN	0.896 ± 0.016	0.973 ± 0.016	0.546 ± 0.136	0.759 ± 0.171	0.188 ± 0.141	0.354 ± 0.325
Transformer	0.983 ± 0.009	0.983 ± 0.009	0.816 ± 0.090	0.816 ± 0.090	0.174 ± 0.103	0.349 ± 0.228

so some LSTM/CNN folds may still benefit from a larger epoch budget or early-stopping study. With three generator seeds, these descriptive metrics do not establish a statistically decisive architecture ranking. All results are synthetic and do not establish performance on real insurance claims. The exact JSON/CSV reports and selected checkpoints are committed in the controlled-comparison results directory; the 25-epoch pilot remains separately archived.

C. Typed-node temporal graph benchmark

The separate custom temporal heterograph encoder was evaluated in a four-mechanism, three-seed synthetic benchmark (12 runs; 2,400 generated claims per run; eight-epoch cap; batch size 64). The chronological split used 55% for training, 10% for model selection, 15% for known-only calibration, and 20% for deployment. Input scaling used the training window only. Each query block read entity summaries aggregated from claims strictly earlier than the block; claims in a query block did not update the entity state seen by other claims in that same block. The custom PyTorch model uses typed Claim, Provider, Hospital, Doctor, Beneficiary, Admission, Package and Diagnosis nodes, with separate provider and referral-target channels. It is not the canonical PyG HGTConv/RGCNConv implementation. Admission is a one-per-claim synthetic proxy and Diagnosis is a severity-bin proxy.

Across all 12 folds, known-class accuracy was 0.954 ± 0.012 and macro-F1 was 0.645 ± 0.107. Novel-mechanism AUROC averaged 0.732 ± 0.170, average precision 0.137 ± 0.152, and recall at the nominal $\alpha = 0.05$ conformal rejection threshold 0.180 ± 0.200. The known false-rejection rate was 0.044 ± 0.009 and the explained-rarity escalation rate was 0.046 ± 0.009. These are descriptive synthetic results; a nominal conformal level does not guarantee coverage under temporal drift.

Table IV shows substantial mechanism dependence. The model separated phantom billing and upcoding more effectively than collusion. Collusion AUROC was 0.494 ± 0.049 and recall was 0.026 ± 0.044, so adding provider and referral-target channels did not solve collusion detection in this generator. Unbundling recall was also low. The explained-rarity gate's novel recall equaled raw conformal novel recall in all 12 runs, indicating that the current gate did not further filter rejected novel claims in this experiment. The aggregate summary does not preserve the denominator for the rare-severe-benign subset, so that metric requires a rerun with explicit counts before interpretation.

The results file and all per-fold reports, predictions and checkpoints are retained in the GitHub Actions artifact linked from the repository's temporal graph benchmark workflow. These findings should not be conflated with the separate provider-level `PotentialFraud` task and do not validate performance on real insurance claims.

VII. LIMITATIONS

All measured experiments reported here use parametric synthetic data, not real claims or claim-level fraud labels from an insurer. Known-class accuracy and novel-mechanism AUROC are distinct tasks and must not be collapsed into one accuracy claim. The standalone LSTM/CNN/Transformer branches use generated proxies and are not fused end-to-end with the temporal graph encoder. The typed-node graph result is a custom PyTorch R-GCN/HGT-style baseline, not the canonical PyG HGTConv/RGCNConv implementation. Admission is represented by one synthetic node per claim, and Diagnosis is a severity-bin proxy rather than an ICD-10 diagnosis node. The graph benchmark is only an initial 12-run evaluation (four held-out mechanisms by three generator seeds, with an eight-epoch cap). Novel recall at the nominal conformal threshold is low on average and varies by mechanism; collusion AUROC is near chance and unbundling recall is weak. The explained-rarity gate's novel recall matched raw conformal recall in all 12 folds, so this benchmark does not demonstrate incremental filtering of those novel cases. The aggregate report does not preserve the denominator for the rare-severe-benign subset; its reported zero false-rejection metric is therefore excluded from interpretation. The v0.2

TABLE IV
MECHANISM-SPECIFIC RESULTS FOR THE TYPED-NODE TEMPORAL GRAPH BASELINE (SYNTHETIC DATA; MEAN ± STANDARD DEVIATION ACROSS THREE GENERATOR SEEDS PER HELD-OUT MECHANISM).

Held-out mechanism	Novel AUROC	Novel AP	Novel recall ($\alpha = 0.05$)
Upcoding	0.835 ± 0.070	0.162 ± 0.033	0.260 ± 0.065
Unbundling	0.714 ± 0.111	0.045 ± 0.017	0.069 ± 0.064
Phantom billing	0.886 ± 0.052	0.314 ± 0.217	0.364 ± 0.310
Collusion	0.494 ± 0.049	0.027 ± 0.009	0.026 ± 0.044

relation-summary pilot uses prior-event summaries. Package rates remain parametric placeholders rather than a reviewed catalogue from the official HBP 2.2 annex, and these reported experiments do not ingest Synthea exports. Synthea itself does not supply claim-level fraud-mechanism labels by default. The conformal guarantee is marginal under exchangeability, while temporal streams may violate this assumption; the explained-rarity gate is outside that guarantee. Manipulable severity evidence may explain away suspicious claims. Broader baselines, ablations, shortcut audits, subgroup/fairness analyses, repeated confirmatory experiments, provenance-checked clinical/HBP reference data, and appropriately labelled real-claim validation remain necessary. Any future deployment must retain human review because incorrect escalation can delay or burden legitimate care.

VIII. Conclusion and Future Work

The repository contains the policy-recommendation prototype, the tabular and relation-summary OW-ERF pilots, independent and controlled LSTM/CNN/Transformer experiments, and a separate typed-node temporal graph baseline. In its first 12-run synthetic benchmark, the graph model achieved known accuracy of 0.954 ± 0.012 and novel AUROC of 0.732 ± 0.170, but novel recall at $\alpha = 0.05$ was only 0.180 ± 0.200. Results depend strongly on mechanism: collusion detection is near chance, and unbundling recall is weak. The current explained-rarity gate did not improve novel recall over conformal rejection in this experiment. These results support continued methodological investigation, not a claim of deployable fraud detection. Next steps include improving graph/label realism, fixing metric denominators, adding baselines and ablations, checking shortcut and subgroup effects, curating source-verified clinical and HBP references, and evaluation on appropriately labelled real claims with human review.

Data and Code Availability

No patient data are used. The synthetic generator, tabular prototype, standalone deep-model benchmark, temporal relation-summary pilot, typed-node temporal graph encoder and evaluation script, Synthea graph adapter, and policy-recommendation prototype are available at https://github.com/ishantpant2121/MUJ-DS-2430010379.
The standalone comparison is reproducible from `scripts/train_evaluate_deep_models.py`; it writes JSON/CSV metrics and 36 model checkpoints to `results/standalone_deep_models/`. Aggregate graph-baseline results are stored under `results/temporal_heterograph_full/`; the full 12-fold run's predictions and model checkpoints are retained in the GitHub Actions artifact. All deep and graph-style results use synthetic data; the separate public provider benchmark uses provider-level `PotentialFraud` labels and is not claim-level mechanism validation.

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